

MARIN XHEMOLLARI

Data Science · University of Michigan · Class of 2028
(248) 719-9978 · marinxhemollari@gmail.com · marinxhemollari.com
github.com/Marin-X · linkedin.com/in/marinxhemollari · Walled Lake, Michigan

SUMMARY

Data Science student at the University of Michigan with a completed Associate degree in Computer Information Systems (Summa Cum Laude, 3.96 GPA). Six independent quantitative finance and data science projects shipped, four with live public demos. Hands-on experience in options pricing, portfolio optimization, risk analytics, backtesting, and data visualization. Background in object-oriented programming, data structures, statistical modeling, and financial data analysis. Seeking internship opportunities in data analytics and quantitative finance.

EDUCATION

University of Michigan, Ann Arbor

B.S. in Data Science, College of Literature, Science & the Arts

Aug 2026 – Apr 2028 (expected) · Ann Arbor, MI

- LSA Transfer Scholarship recipient (awarded on admission, 2026). Transferred with 62 credit hours accepted.
- Fall 2026 coursework: EECS 280 (Programming and Data Structures), MATH 205 (Multivariable Calculus), ECON 101 (Principles of Economics), PHIL 380 (Philosophical Perspectives).

Oakland Community College

Associate in Applied Science, Computer Information Systems (Software Engineering)

Aug 2023 – Dec 2025 · Farmington Hills, MI

- Graduated Summa Cum Laude, cumulative GPA 3.96 / 4.00. Dean's List, multiple consecutive terms.
- Coursework: Data Structures, Object-Oriented Programming (Java, C++), Software Engineering, Database Systems (MySQL), Web Development (PHP, JavaScript), Mobile Development (Android / Kotlin), Calculus I & II, Discrete Structures, Engineering Physics I & II.

TECHNICAL SKILLS

Programming:	Java, C++, Python, SQL, JavaScript, Kotlin, HTML/CSS, PHP
Libraries & Tools:	NumPy, pandas, SciPy, Matplotlib, Streamlit, Git, GitHub, IntelliJ IDEA, Android Studio, MySQL
Finance & Analytics:	Options Pricing (Black-Scholes), Portfolio Optimization, Risk Analytics (VaR, CVaR), Backtesting, Statistical Modeling, Financial Data Analysis
Spoken Languages:	Albanian (native), English (fluent), Romanian (intermediate)

SELECTED PROJECTS

NewsLens — News-Sentiment Alpha Factor Research

Python, FinBERT, pandas, Streamlit · newslens-marinx.streamlit.app

- Built end-to-end sentiment-to-alpha research pipeline scoring news headlines for all 500 S&P 500 constituents using FinBERT transformer embeddings.
- Backtested 65 long-short portfolio configurations. Achieved Information Coefficient t-statistic of +2.98, Sharpe ratio of 2.63, and 83% directional hit rate.

Probatus — Event-Driven Multi-Strategy Backtester

Python, pandas, NumPy, Streamlit · probatus-marinx.streamlit.app

- Developed event-driven backtesting framework simulating five trading strategies with IBKR commission schedules and Almgren-Chriss market-impact modeling.
- Implemented walk-forward validation, parameter-sweep heatmaps, and equity-curve diagnostics for overfitting detection.

Derivexus — Options Pricing & Implied-Volatility Engine

Python, NumPy, SciPy, Streamlit · derivexus-marinx.streamlit.app

- Implemented Black-Scholes-Merton valuation, full Greek computation (delta, gamma, theta, vega, rho), and Brent-method implied-volatility solver.
- Constructed 3D implied-volatility surfaces with quadratic smile fitting. Built mispricing scanner for systematic divergence detection.

QuantFolio — Portfolio Construction & Risk Analytics

Python, SciPy, pandas, Streamlit · quantfolio-marinx.streamlit.app

- Built interactive tool computing efficient frontier, Max Sharpe allocation, and Risk Parity weighting with Fama-French three-factor and five-factor regression.
- Implemented Monte Carlo simulation for VaR and CVaR tail-risk estimation with live market data.

AlphaAgents — Multi-Agent AI Hedge-Fund Simulator

Python, Anthropic API, yfinance, Streamlit

- Architected multi-agent portfolio management system: four analyst agents (fundamentals, technicals, macro, sentiment) coordinated by a Portfolio Manager meta-agent reasoning over live market data.

DepthLens — Real-Time Order-Book Viewer (Android)

Kotlin, Android, MVVM, OkHttp, SQLite

- Built native Android application streaming Binance Level 2 order-book data over WebSocket with real-time VWAP, TWAP, and Order Flow Imbalance analytics.

WORK EXPERIENCE

Youri Sushi — Server

Sep 2022 – Present · Novi, MI

- Managed high-volume table service while completing full-time technical coursework and shipping six independent software projects.
- Four years of continuous employment through two degree programs, demonstrating sustained reliability and time management under competing demands.

HONORS & AWARDS

- Summa Cum Laude, Oakland Community College (December 2025).
- LSA Transfer Scholarship, University of Michigan (2026).
- Dean's List, Oakland Community College (multiple consecutive terms).